

dhc-pipeline: a hardened-image catalogue

A miniature Docker Hardened Images catalogue of 6 images, 4 Helm charts and a CVE triage lane, built to a three-day delivery plan on the real DHI toolchain, then left operating.

What it is

dhc-pipeline reproduces the working surface of a DHI engineering role end to end: declarative image definitions, automated builds with supply-chain attestations, upstream release tracking, Helm chart adaptation onto non-root images, Go integration tests against real Kubernetes, and CVE triage recorded as portable, machine-readable decisions. Figure 1 overleaf is the whole machine on one diagram.

Nothing is simulated: images are authored in **native DHI syntax** and built by Docker's own `dhc.io/build` BuildKit frontend, after a three-hour spike proved it works outside Docker's infrastructure on the Community tier (ADR 0001). Every other layer is equally standard: Renovate, Trivy and Gripe, OpenVEX/vexctl, cosign, Syft, Kyverno, Ginkgo v2 on kind.

Delivered

6 images, 4 archetypes	from-source · monorepo to 3 images · tarball repackaging · C-source stateful
4 charts	3 upstream adaptations consumed unmodified, 1 owned; every deviation documented with its rationale
6 CI workflows	validate · build and release · chart gate · kind e2e · Renovate cron (≤6h) · daily rescan
~2,600 lines of tested Go	the kind e2e harness and the rescan reporter, both written test-first
28 PRs, 109 commits	against 8 EARS-notated requirements, 2 ADRs and a dated triage log

Every published image carries a cosign keyless signature, an SPDX SBOM and SLSA provenance, plus an OpenVEX attestation wherever a triage decision exists. Kyverno gates rendered manifests on digest pinning, allowed registries and non-root execution. The e2e suite asserts that live pods run as UID 65532 with a read-only root filesystem and all capabilities dropped, then probes them functionally: cert-manager issues a Certificate, Grafana answers health.

Scan findings become recorded decisions rather than anecdotes. Figure 2 overleaf is the model that governs them: a single question decides whether a finding applies to the artifact at all, and the four treatments underneath it each produce a different, auditable artifact.

What running it for real turned up

- **An upstream artifact that moves.** Grafana rewrites its `/oss/release/` tarballs about twenty hours after a release — all four objects of the 2026-07-21 batch inside one six-minute window, so this is routine automation rather than a one-off. A digest pinned on release day silently stops matching, with nothing in the version to explain it. Our own build provenance proved the original pin had verified on both architectures, which is what ruled out a bad pin on our side. Fixed by pinning the build-scoped artifact, which has not moved since release day, and by adding a CI check that fetches every pinned architecture so an unexercised pin cannot enter again.
- **An invisible class of security release.** Grafana ships out-of-band fixes as semver build metadata (`13.0.1+security-01`), which semver ignores for precedence, so it never ranks *above* the plain release. The tracker would have reported us up to date straight through a security release on the one image carrying every open CVE.
- **Bumping is not monotonic.** The 13.0.4 to 13.1.1 fix bump cleared three stdlib findings from the main binary and introduced four new ones inside a bundled plugin. A version bump swaps a dependency graph rather than shrinking it.

• **Status: operating**

Renovate opens bump PRs on a four-hour cron, the daily rescan has filed eleven CVE issues against nine distinct findings, enriched with EPSS and KEV, and the grafana gate stays red on fifteen HIGH and one CRITICAL, including the one where the honest answer is still *we do not know yet*.

Figure 1: the pipeline

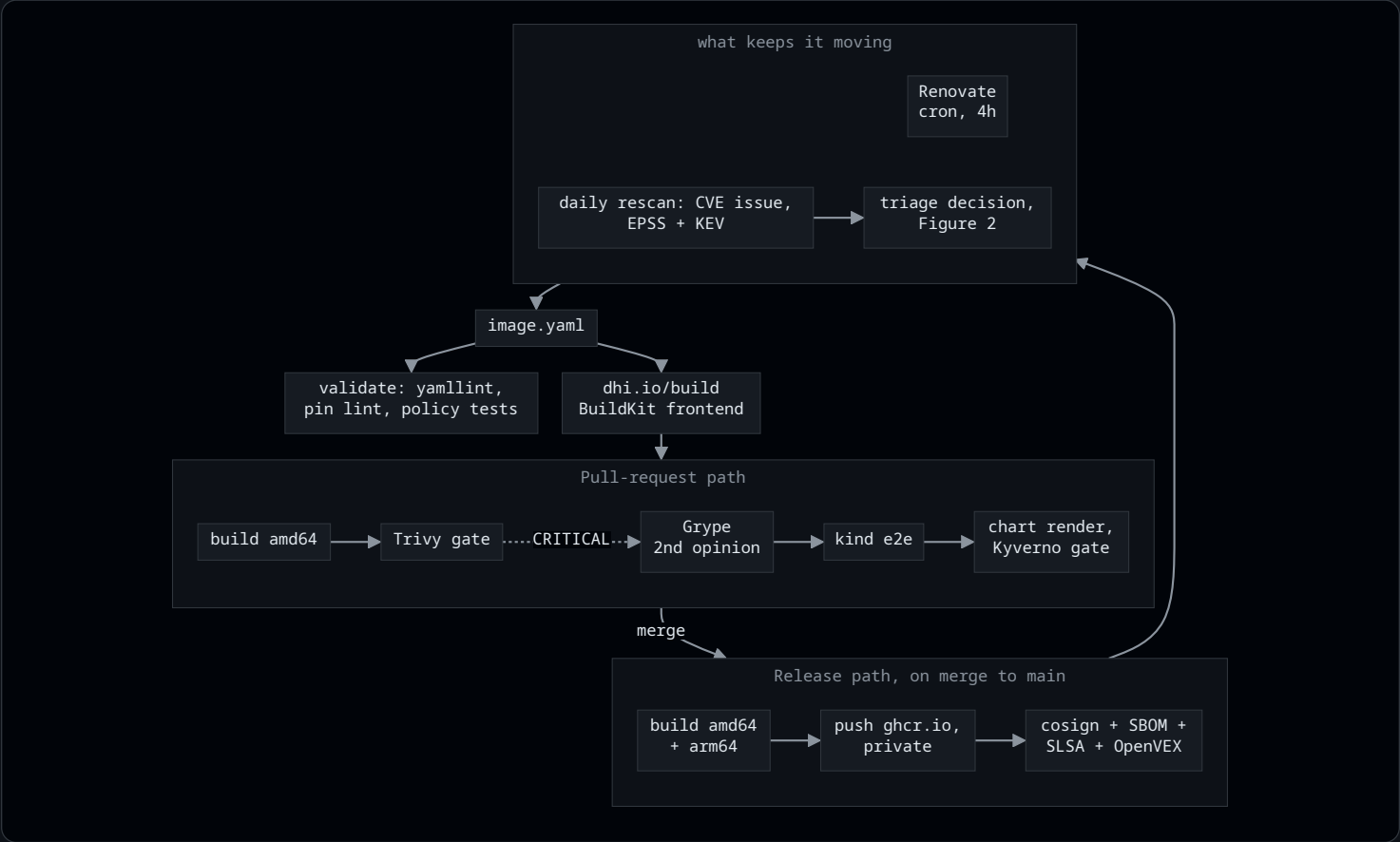


Figure 1. Nothing reaches the registry unsigned, uninventoried or unattested, and nothing stays current by hand.

Figure 2: the triage decision model

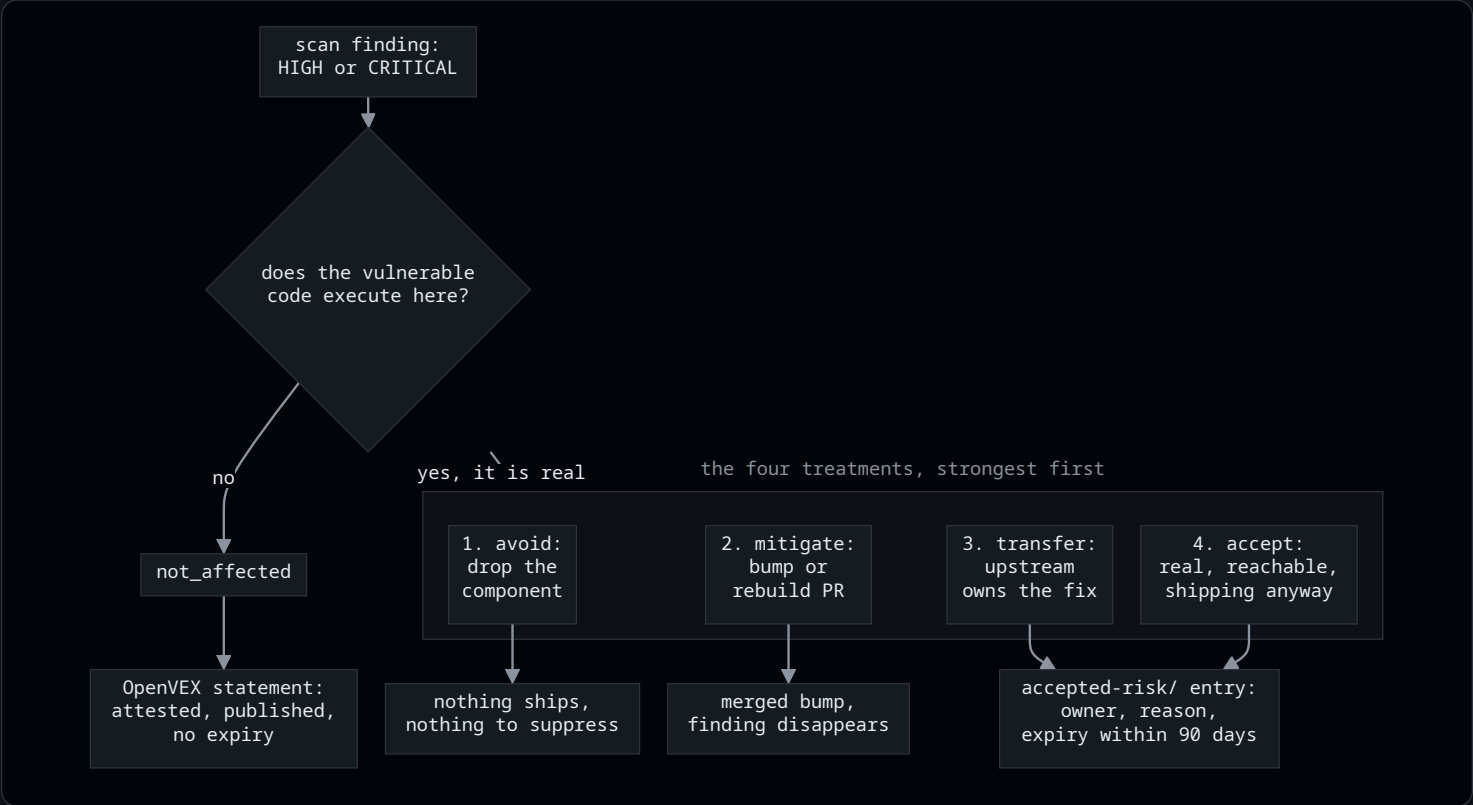


Figure 2. Two different questions. VEX says *does this apply to the artifact?* and is attested to the image with no expiry. Accepted-risk says *what are we doing about our exposure?* and is internal, never attested, and expiring. Conflating them is VEX-washing, so the pipeline keeps them in separate files with separate lifetimes.